Chromosomes - super packets - contains genes

X chromosomes have ~ 1000s of genes
Y chromosomes now have ~50-70 genes

Metabollic functions, neural development, repair, & many more of the genes specific to females & the genes required for every organism is retained in the X chromosome.

The Y chromosome packet - has mostly only the genes required for specifically male functions - hardly 20 genes express outside the purview of testes.

Females have 2 X chromosomes & males 1 X & 1 Y

So females have extra copy of every gene - which makes the XX type more stable & resilient - if 1 copy of a gene is damaged, it has another copy to fall back on.

Males lack this resilience - If a gene in either of their X or Y chromosome is damaged - there is no backup.

One of the manifestations of this is the skew of neurodevelopmental disorders in males.

Here begins the biological, neurobiological, behavioural differences in the genders as determined by genetics.

Y chromosome has the SRY genes - expressed only in males obviously - this gene influences testes formation - this is the switch that shifts the other default female development of brain & body to male development. With the SRY gene activated, testosterone surges in the brain , which is necessary to alter the architecture of the development of the brain to male setting , instead of the default female plan. Without this, an organism with XY chromosome's brain wouldn't be fully formed.

The other set of influential gene contributing to sexual differences is the Androgen Receptor genes - this AR gene expression differs by sex, but also by individual. But largely, statistically, AR gene density is higher in males. AR gene receptors work with the testosterone - so it is not just how much testosterone there is that determines male development, brain structures and therefore behaviors, etc - but it is the combination of both the testosterone level & the AR gene density and the sensitivity of the receptors which also varies over a range.
This AR gene sits in the X chromosome.

The SRY gene in the Y chromosome triggers testosterone surge, which activates the AR gene, proportional to their density & sensitivity across different cells & brain regions.

AR activation is particularly high in brain regions like amygdala, hypothalamus, basal ganglia, nucleus accumbens - the reward circuit. AR activations in these regions have its effects -

Amygdala

AR activation:

- Increases dendritic growth in threat circuits
- Enhances reactivity to dominance challenges
- Reduces fear inhibition under competition

Result

- Faster aggression response
- Greater territorial behavior

Hypothalamus

AR activation:

- Expands certain nuclei (e.g., sexually dimorphic nucleus)
- Strengthens mating and dominance drives

Result:

- Higher drive for rank positioning
- Competitive hierarchy sensitivity

Reward circuits (nucleus accumbens / basal ganglia)

AR activation:

- Enhances dopamine signaling
- Increases reward sensitivity for status, risk & sexual novelty

Result:

- Stronger risk loops
- Stronger pursuit behaviors

The next set of genes that contribute to sexual differences is the Estrogen Receptors present in the X chromosome - which also both males and females have and these receptors bind with the estradiol - which is also produced in both genders. They modulate gene transcription to influence development, long term plasticity and also contribute to fast effects in the synapses, mitochondria, dendritic spines & cell membrane, causing more short term fluctuations in the body, moods and more- in both genders.

Females show higher baseline ER expression in certain brain regions - hypothalamus, PFC, hippocampus & amygdala - this expression also fluctuates across the menstrual cycle.